8.5 Integration of Rational Functions by Partial Fractions

A *rational function* is a function of the form $\frac{f(x)}{g(x)}$, where $f(x)$ and $g(x)$ are polynomials with $g(x) \neq 0$.

We say a rational function is *proper* if $\deg(f) < \deg(g)$, and *improper if* $\deg(f) \geq \deg(g)$.

e.g. $\frac{x^2}{x^3+1}$ proper, $\frac{x^3+1}{x^2+4}$ improper

If we have an improper rational function to integrate, we can first apply long division to change it to a proper rational function. For example,

$$\int \underbrace{\frac{x^3+3x+1}{x^2+1}}_{\text{improper}} = \int x + \underbrace{\frac{2x+1}{x^2+1}}_{\text{proper}} dx = \frac{x^2}{2} + \int \frac{2x}{x^2+1}dx + \int \frac{1}{x^2+1}dx$$

$= \ln(x^2+1)$ $= \ln|x^2+1|$ $= \tan^{-1}(x)$

To integrate a proper rational function, we try to express it as a sum of partial fractions (if we can), e.g.

$$\int \frac{5x-3}{x^2-2x-3}dx = \int \frac{2}{x+1} + \frac{3}{x-3}dx$$

$$= 2\ln|x+1| + 3\ln|x-3| + C$$

General Description of the Method

e.g. $\frac{x^2+4}{x^2-x+1} = 1 + \frac{x+3}{x^2-x+1}$ (improper / proper)

Success in writing a rational function $\frac{f(x)}{g(x)}$ as a sum of partial fractions depends on two things:

- *The degree of $f(x)$ must be less than the degree of $g(x)$*. That is, the fraction must be proper. If it isn't, divide $f(x)$ by $g(x)$ and work with the remainder term.
- *We must know the factors of $g(x)$*. In theory, any polynomial with real coefficients can be written as a product of real linear factors and real quadratic factors. In practice, the factors may be hard to find.

Remark: We say a quadratic polynomial is *irreducible* if it is not a product of two (real) linear factors. That is, it has no real root. For example, $x^2 + x + 1$ is an irreducible polynomial, but $x^2 + 3x + 2$ is reducible, since $x^2 + 3x + 2 = (x + 1)(x + 2)$.

ax^2+bx+c is irreducible $\Leftrightarrow b^2-4ac<0$

$$\frac{f(x)}{g(x)} = \frac{f(x)}{(x-r_1)^{\alpha_1}(x-r_2)^{\alpha_2}\cdots(x-r_M)^{\alpha_M}(x^2+b_1x+c_1)^{\beta_1}\cdots(x^2+b_Nx+c_N)^{\beta_N}} = \frac{A}{x-r_1}+\frac{B}{(x-r_1)^2}+\cdots + \frac{Dx+E}{x^2+b_1x+c_1}+\cdots$$

Methods of Partial Fractions ($f(x)/g(x)$Proper)

1. Let $x - r$ be a linear factor of $g(x)$. Suppose that $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$. Then, to this factor, assign the sum of the m partial fractions:

$$\frac{f(x)}{(x-r)^m} = \frac{A_1}{(x-r)} + \frac{A_2}{(x-r)^2} + \ldots + \frac{A_m}{(x-r)^m}.$$

Do this for each distinct linear factor of $g(x)$.

e.g. $\frac{x}{(x+2)^2(x+3)^3} = \frac{A}{(x+2)} + \frac{B}{(x+2)^2} + \frac{C}{x+3} + \frac{D}{(x+3)^2} + \frac{E}{(x+3)^3}$

2. Let $x^2 + px + q$ be an irreducible quadratic factor of $g(x)$ so that $x^2 + px + q$ has no real roots. Suppose that $(x^2 + px + q)^n$ is the highest power of this factor that divides $g(x)$. Then, to this factor, assign the sum of the n partial fractions:

$$\frac{B_1x+C_1}{(x^2+px+q)} + \frac{B_2x+C_2}{(x^2+px+q)^2} + \ldots + \frac{B_nx+C_n}{(x^2+px+q)^n}.$$

Do this for each distinct quadratic factor of $g(x)$.

$\frac{x^2+1}{(x^2+x+1)^2(x^2+4)(x-1)^2} = \frac{Ax+B}{x^2+x+1} + \frac{Cx+D}{(x^2+x+1)^2} + \frac{Ex+F}{x^2+4} + \frac{G}{x-1} + \frac{H}{(x-1)^2}$.

3. Set the original fraction $f(x)/g(x)$ equal to the sum of all these partial fractions. Clear the resulting equation of fractions and arrange the terms in decreasing power of x.

4. Equate the coefficients of corresponding powers of x and solve the resulting equations for the undetermined coefficients.

Example:

1. $\frac{2x+1}{(x^2+3x+2)^2} = \frac{2x+1}{(x+1)^2(x+2)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+2)} + \frac{D}{(x+2)^2}$

can be factorised

$= (x+1)(x+2)$

To find A, B, C, D, multiply both sides by $(x+1)^2(x+2)^2$ to get

$$2x+1 = A(x+1)(x+2)^2 + B(x+2)^2 + C(x+1)^2(x+2) + D(x+1)^2$$

Compare coefs: $x^3 \rightarrow 0 = A + C$

2. $\dfrac{x^2+2x+1}{(x^2+x+1)^2} = \dfrac{Ax+B}{x^2+x+1} + \dfrac{Cx+D}{(x^2+x+1)^2} = \dfrac{1}{x^2+x+1} + \dfrac{x}{(x^2+x+1)^2}$

To find A, B, C, D, multiply by $(x^2+x+1)^2$ to get

$$x^2+2x+1 = (\overset{0}{\cancel{A}}x+\overset{1}{\cancel{B}})(x^2+x+1) + \overset{1}{\cancel{C}}x+D .$$

Compare coefs to solve for A, B, C, D.

x^3 : $\quad 0 = A$

x^2 : $\quad 1 = B$

x : $\quad 2 = B + C = 1 + C \Rightarrow C = 1$

x^0 : $\quad 1 = B + D = 1 + D \Rightarrow D = 0$

Example Distinct Linear Factors

Evaluate $\int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)}dx$ using partial fractions.

$$\frac{x^2+4x+1}{(x-1)(x+1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{x+3}$$

$$\Rightarrow \quad x^2+4x+1 = A(x+1)(x+3) + B(x-1)(x+3) + C(x-1)(x+1)$$

$$\Rightarrow \quad x^2+4x+1 = (A+B+C)x^2 + (4A+2B)x + (3A-3B-C)$$

comparing coefs, we have

$$\left.\begin{array}{ll} x^2: & A+B+C=1 \\ x^1: & 4A+2B=4 \\ x^0: & 3A-3B-C=1 \end{array}\right\} \Rightarrow \begin{array}{l} A = 3/4 \\ B = 1/2 \\ C = -1/4 \end{array}$$

$$\therefore \int \frac{x^2+4x+1}{(x-1)(x+1)(x+3)}dx = \int \frac{3/4}{x-1} + \frac{1/2}{x+1} - \frac{1/4}{x+3}dx = \frac{3}{4}\ln|x-1| + \frac{1}{2}\ln|x+1| - \frac{1}{4}\ln|x+3| + C$$

Example A Repeated Linear Factor

Evaluate $\int \frac{6x+7}{(x+2)^2} dx$.

$$\frac{6x+7}{(x+2)^2} = \frac{A}{x+2} + \frac{B}{(x+2)^2}$$

$$\Rightarrow \quad 6x+7 = A(x+2)+B$$

$$= Ax + (2A+B)$$

Compare coefs $\Rightarrow$ $A = 6$ & $7 = 2A+B = 12+B \Rightarrow B = -5$

$$\int \frac{6x+7}{(x+2)^2} dx = \int \frac{6}{x+2} - \frac{5}{(x+2)^2} dx = 6\ln|x+2| + \frac{5}{x+2} + C$$

Example Integrating an Improper Fraction

Evaluate $\int \frac{2x^3-4x^2-x-3}{x^2-2x-3}dx$. ← improper

$$x^2-2x-3 \;\overline{)\; 2x^3-4x^2-x-3}$$

$$2x$$

$$2x^3-4x^2-6x$$

$$5x-3$$

$$\Rightarrow \frac{2x^3-4x^2-x-3}{x^2-2x-3} = 2x + \frac{5x-3}{x^2-2x-3}$$

$$= 2x + \frac{5x-3}{(x+1)(x-3)}$$

$$\frac{5x-3}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$$

$$= 2x + \frac{2}{x+1} + \frac{3}{x-3}$$

$$\ldots\ldots \quad A = 2, \quad B = 3$$

$$\therefore \int \frac{2x^3-4x^2-x-3}{x^2-2x-3}dx = \int 2x + \frac{2}{x+1} + \frac{3}{x-3}\,dx$$

$$= x^2 + 2\ln|x+1| + 3\ln|x-3| + C$$

Example Integrating with an Irreducible Quadratic Factor in the Denominator

Evaluate $\int \frac{-2x+4}{(x^2+1)(x-1)^2}\,dx$.

irreducible

$$\frac{-2x+4}{(x^2+1)(x-1)^2} = \frac{Ax+B}{x^2+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2}$$

Multiplying by $(x^2+1)(x-1)^2$, we get

$$-2x+4 = (Ax+B)(x-1)^2 + C(x^2+1)(x-1) + D(x^2+1)$$

$$= (A+C)x^3 + (-2A+B-C+D)x^2 + (A-2B+C)x + (B-C+D)$$

Compare coefs to get

x^3 : $A + C = 0$

x^2 : $-2A + B - C + D = 0$

x^1 : $A - 2B + C = -2$

x^0 : $B - C + D = 4$

$A = 2$, $B = 1$, $C = -2$, $D = 1$

$$\int \frac{-2x+4}{(x^2+1)(x-1)^2}\,dx = \int \frac{2x+1}{x^2+1} - \frac{2}{x-1} + \frac{1}{(x-1)^2}\,dx$$

$$= \int \frac{2x}{x^2+1}\,dx + \int \frac{1}{x^2+1}\,dx - 2\ln|x-1| - \frac{1}{x-1}$$

$$= \ln|x^2+1| + \tan^{-1}(x) - 2\ln|x-1| - \frac{1}{x-1} + C$$

$u = x^2+1$

Example A Repeated Irreducible Quadratic Factor

Evaluate $\int \frac{1}{x\,(x^2+1)^2}\,dx$.

irreducible

$$\frac{1}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2}$$

$$1 = A(x^2+1)^2 + (Bx+C)x(x^2+1) + (Dx+E)x$$

$$1 = (A+B)x^4 + Cx^3 + (2A+B+D)x^2 + (C+E)x + A$$

$$\begin{aligned} A+B &= 0 \\ C &= 0 \\ 2A+B+D &= 0 \\ C+E &= 0 \\ A &= 1 \end{aligned}$$

$$\begin{aligned} A &= 1 \\ B &= -1 \\ C &= 0 \\ D &= -1 \\ E &= 0 \end{aligned}$$

$$\int \frac{1}{x(x^2+1)}\,dx = \int \frac{1}{x} - \frac{x}{x^2+1} - \frac{x}{(x^2+1)^2}\,dx$$

$u = x^2+1$

$$= \ln|x| - \frac{1}{2}\ln|x^2+1| - \frac{1}{2(x^2+1)} + C$$

$u = x^2+1$

$du = 2x\,dx$

The Heaviside "Cover-up" Method for Linear Factors

Example Find A, B, and C in the partial fraction expansion

$$\frac{x^2+1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}.$$

① Multiply both sides with $(x-1)$ to get

$$\frac{x^2+1}{(x-2)(x-3)} = A + \frac{B(x-1)}{x-2} + \frac{C(x-1)}{x-3}$$

Setting $x=1$, we get $\frac{1+1}{(1-2)(1-3)} = A \Rightarrow A = 1$

② Multiply by $(x-2)$ instead, we get

$$\frac{x^2+1}{(x-1)(x-3)} = \frac{A(x-2)}{x-1} + B + \frac{C(x-2)}{x-3}$$

Setting $x=2$, we get $\frac{5}{-1} = B \Rightarrow B = -5$.

③ Similarly, "cover-up" $(x-3)$ to get $C \Rightarrow C = \frac{3^2+1}{(3-1)(3-2)} = 5$

The Heaviside "Cover-up" Method for Linear Factors

Quick way for when $g(x) = (x - r_1)(x - r_2)\ldots(x - r_n)$, and r_i distinct.

Heaviside Method

1. Write the quotient with $g(x)$ factored:

$$\frac{f(x)}{g(x)} = \frac{f(x)}{(x-r_1)(x-r_2)\ldots(x-r_n)} = \frac{A_1}{x-r_1} + \frac{A_2}{x-r_2} + \ldots + \frac{A_n}{x-r_n}.$$

2. Cover the factors $(x - r_i)$ of $g(x)$ one at a time, each time replacing all the uncovered x's by the number r_i. This gives a number A_i for each root r_i:

$$A_1 = \frac{f(r_1)}{(r_1-r_2)\ldots(r_1-r_n)}$$

$$A_2 = \frac{f(r_2)}{(r_2-r_1)(r_2-r_3)\ldots(r_2-r_n)} \quad \ldots\ldots$$

$$A_n = \frac{f(r_1)}{(r_n-r_1)(r_n-r_2)\ldots(r_n-r_{n-1})}.$$

3. Write the partial fraction expansion of $f(x)/g(x)$ as

$$\frac{f(x)}{g(x)} = \frac{A_1}{(x-r_1)} + \frac{A_2}{(x-r_2)} + \ldots + \frac{A_n}{(x-r_n)}.$$

Example Integrating with the Heaviside Method

Evaluate $\int \frac{x+4}{x^3+3x^2-10x}\,dx.$

$$\frac{x+4}{x^3+3x^2-10x} = \frac{x+4}{x(x-2)(x+5)} = \frac{A}{x} + \frac{B}{x-2} + \frac{C}{x+5}$$

$$A = \frac{0+4}{(0-2)(0+5)} = -\frac{2}{5} \qquad \text{(cover "x", replace } x=0\text{)}$$

$$B = \frac{2+4}{2(2+5)} = \frac{3}{7}. \qquad \text{(cover "x-2", set } x=2\text{)}$$

$$C = \frac{-5+4}{(-5)(-5-2)} = -\frac{1}{35} \qquad \text{(cover "x+5", set } x=-5\text{)}$$

$$\therefore \int \frac{x+4}{x^3+3x^2-10x}\,dx = \int -\frac{2}{5}\cdot\frac{1}{x} + \frac{3}{7}\frac{1}{x-2} - \frac{1}{35}\frac{1}{x+5}\,dx$$

$$= -\frac{2}{5}\ln|x| + \frac{3}{7}\ln|x-2| - \frac{1}{35}\ln|x+5| + C$$

Other Ways to Determine the Coefficients

Using Differentiation: Find A, B, and C in the equation

$$\frac{x-1}{(x+1)^3} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3}.$$

$= \frac{1}{(x+1)^2} - \frac{2}{(x+1)^3}$

$\Rightarrow \quad x-1 = A(x+1)^2 + B(x+1) + C$

set $x = -1$ to get $\quad \underline{-2 = C}$

So $\quad x-1 = A(x+1)^2 + B(x+1) - 2$

differentiate wrt x we get

$$1 = 2A(x+1) + B$$

set $x = -1$ to get $\quad \underline{1 = B}$

So $\quad 1 = 2A(x+1) + 1$

diff wrt $x \quad 0 = 2A \Rightarrow \underline{A = 0}$

Assigning Numerical Value to x:

Find A, B, and C in $\dfrac{x^2+1}{(x-1)(x-2)(x-3)} = \dfrac{A}{x-1} + \dfrac{B}{x-2} + \dfrac{C}{x-3}$.

$\Rightarrow \quad x^2+1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$

Set $x=1 \Rightarrow 2 = A(-1)(-2) \Rightarrow A = 1$

Set $x=2 \Rightarrow 5 = B(1)(-1) \Rightarrow B = -5$

Set $x=3 \Rightarrow 10 = C(2)(1) \Rightarrow C = 5$

$$\Rightarrow \frac{x^2+1}{(x-1)(x-2)(x-3)} = \frac{1}{x-1} - \frac{5}{x-2} + \frac{5}{x-3}$$

Exercise

Evaluate $\int \frac{2x+2}{(x^2+1)(x-1)^3}dx.$

1. $\int \frac{3x+11}{x^2-x-6}\,dx$

2. $\int \frac{x}{(x-1)^2(x+2)}\,dx$

3. $\int \frac{2x^2-x+4}{x^3+4x}\,dx$

4. $\int \frac{x^3+x}{x-1}\,dx$

5. $\int \frac{x^3+2x+1}{(x^2+1)^2}\,dx$

$$\frac{2x+2}{(x^2+1)(x-1)^3} = \frac{Ax+B}{x^2+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2} + \frac{E}{(x-1)^3}$$

irreducible

Multiply both sides with $(x^2+1)(x-1)^3$ to get

$$2x+2 = (Ax+B)(x-1)^3 + C(x^2+1)(x-1)^2 + D(x^2+1)(x-1) + E(x^2+1)$$

Set $x=1$ to get $4 = 2E \Rightarrow E = 2$

Compare coefs on both sides, we have

x^4 : $A + C = 0 \Rightarrow C = 0$

x^3 : $-3A + B - 2C + D = 0 \Rightarrow B + D = 0$

x^2 : $3A - 3B + 2C - D + 2 = 0$

x : $-A + 3B - 2C + D = 2$

$\Rightarrow 2A + 2 = 2 \Rightarrow A = 0$

~~x^0 : $-B + C - D + 2 = 2$~~

$3B + D = 2 \Rightarrow 2B = 2 \Rightarrow B = 1 \Rightarrow D = -1$

$$\int \frac{2x+2}{(x^2+1)(x-1)^3}\,dx = \int \frac{1}{x^2+1} - \frac{1}{(x-1)^2} + \frac{2}{(x-1)^3}\,dx$$

$$= \tan^{-1}(x) + \frac{1}{x-1} - \frac{1}{(x-1)^2} + C$$